

- 16 (a) Use the iteration formula  $x_{n+1} = \sqrt[3]{10 - 2x_n}$  to find the values of  $x_1$ ,  $x_2$  and  $x_3$   
Start with  $x_0 = 2$

$$x_1 = \dots\dots\dots$$

$$x_2 = \dots\dots\dots$$

$$x_3 = \dots\dots\dots$$

(3)

The values of  $x_1$ ,  $x_2$  and  $x_3$  found in part (a) are estimates of the solution of an equation of the form  $x^3 + ax + b = 0$  where  $a$  and  $b$  are integers.

- (b) Find the value of  $a$  and the value of  $b$ .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(1)

(Total for Question 16 is 4 marks)